

CAPSTONE PROJECT REPORT



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Library Management System

1. Introduction

The Library Management System is a full-stack web-based application designed to provide an efficient and user-friendly platform for managing library operations. In traditional libraries, managing records manually can be time-consuming, error-prone, and inefficient. This project aims to digitize and automate core library functionalities such as book management, issuing, returning, and reservation.

The application allows users to browse a collection of books categorized by authors and genres. It provides features such as searching books, issuing available books, reserving unavailable books, and viewing personal records like issued and reserved books. Users can register and log in to access personalized features, making the system more interactive and user-centric.

From a technical perspective, the system follows a **three-tier architecture**, consisting of frontend, backend, and database layers. The frontend is developed using HTML, CSS, and JavaScript to provide a responsive and interactive user interface. The backend is implemented using Node.js and Express.js, which handle API requests, business logic, and communication with the database. The database is built using SQL, ensuring efficient data storage through structured relational tables.

One of the key objectives of this project is to demonstrate the practical implementation of Database Management System (DBMS) concepts such as normalization, relationships, indexing, and query optimization. The system maintains multiple related tables such as users, books, authors, genres, and reservations, ensuring data consistency and integrity.

Additionally, the project emphasizes user experience by incorporating a clean and modern UI design. Features like a card-based book layout, search filters, and intuitive navigation contribute to a smooth and engaging interface.

Overall, this Library Management System demonstrates full-stack development by integrating frontend design, backend logic, and database management into a unified system. It fulfills academic requirements and provides a strong foundation for building more advanced digital library systems.

2. System Architecture

The Library Management System follows a **three-tier architecture**, which separates the application into:

- Presentation Layer (Frontend)
- Application Layer (Backend)
- Data Layer (Database)

2.1 Overview of Architecture

Flow of system:

User → Frontend → Backend → Database → Backend → Frontend → User

2.2 Presentation Layer (Frontend)

The frontend layer is responsible for user interaction and displaying data.

Technologies Used:

- HTML
- CSS
- JavaScript

Key Responsibilities:

- Display books in card format
- Provide search and filter functionality
- Handle user actions (issue, reserve, view)
- Send API requests using Fetch API

Important Components:

- Login/Register Page
- Dashboard
- Browse Books Page
- Reservations Page
- Notifications Section

2.3 Application Layer (Backend)

The backend processes requests and contains business logic.

Technologies Used:

- Node.js
- Express.js

Key Responsibilities:

- Manage book issuing and reservation logic
- Authenticate users
- Send data in JSON format

Main API Endpoints:

- `/books` → Fetch books
- `/authors` → Manage authors
- `/genres` → Manage genres
- `/reservations` → Handle reservations
- `/issue` → Issue books
- `/return` → Return books
- `/login` → User login
- `/register` → User registration

2.4 Data Layer (Database)

The database stores structured data.

Technology Used:

- SQL (MySQL/PostgreSQL)

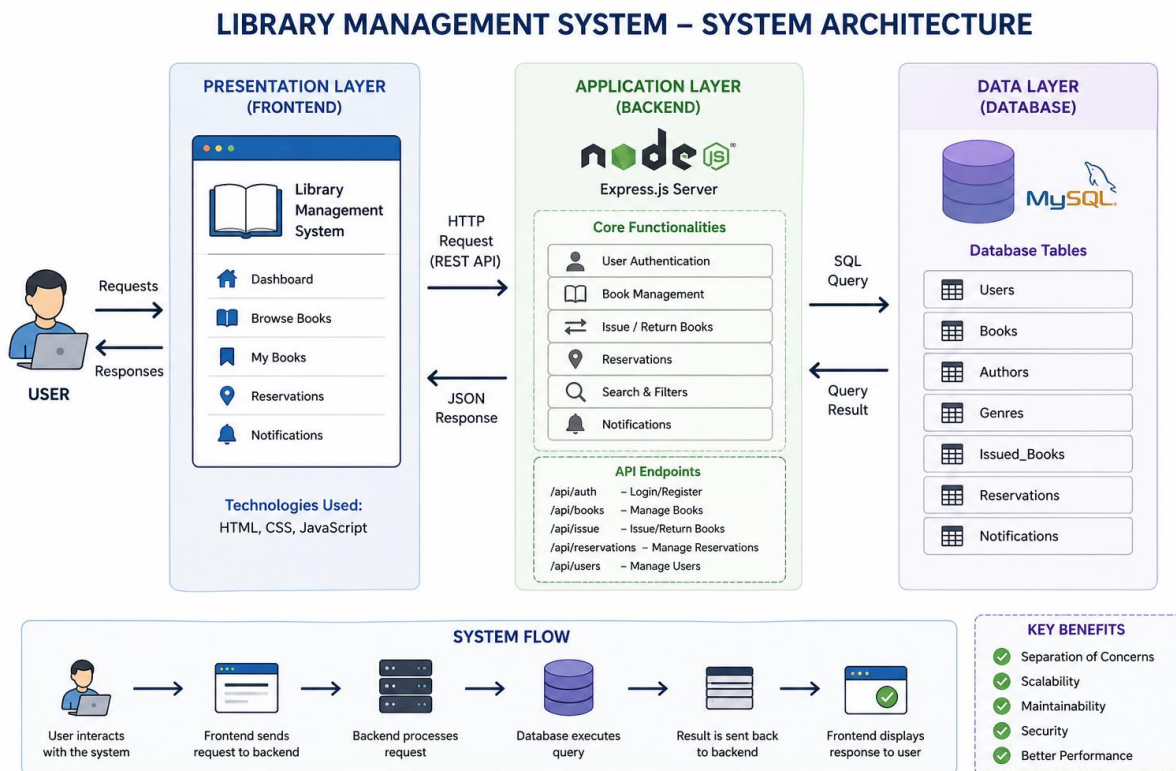
Key Responsibilities:

- Store books, users, and transactions
- Maintain relationships using foreign keys
- Ensure data integrity

2.5 Data Flow Explanation

1. User interacts with UI
2. Request sent to backend
3. Backend queries database
4. Database returns result
5. Backend sends response
6. UI updates dynamically

2.6 System Architecture



2.7 Advantages of Architecture

- Separation of concerns
- Scalability
- Maintainability
- Security
- Efficient data handling

3. OBJECTIVE

- To automate library operations
- To implement CRUD operations using APIs
- To design a relational database
- To provide a user-friendly interface
- To implement issue and reservation system

4. DATABASE DESIGN

Books

- book_id (Primary Key)
- title
- author_id (Foreign Key)
- genre_id (Foreign Key)
- total_copies
- available_copies
- image_url

Authors

- author_id
- name

Genres

- genre_id
- name

Users

- user_id
- username
- password

Reservations

- reservation_id
- user_id
- book_id
- status

Issued Books

- issue_id
- user_id
- book_id
- issue_date
- due_date

5. SEARCH FEATURES

The system supports:

- Search by book title
- Search by author
- Search by genre

Implemented using SQL JOIN queries.

6. ANALYTICAL QUERIES

- Total number of books
- Books available vs issued
- Most popular genre
- Number of books per author

7. FRONTEND DESIGN

Includes:

- Dashboard
- Book catalog page
- Reservation page

Features:

- Search bar
- Book cards
- Issue & Reserve buttons
- Filters

8. USER INTERFACE

- Sidebar navigation
- Card-based layout
- Responsive design
- Hover effects
- Clean UI

9. AUTHENTICATION SYSTEM

Users can:

- Register
- Login

Authentication is handled using database validation.

10. SYSTEM FUNCTIONALITY

Issue Book:

- If available → Issue

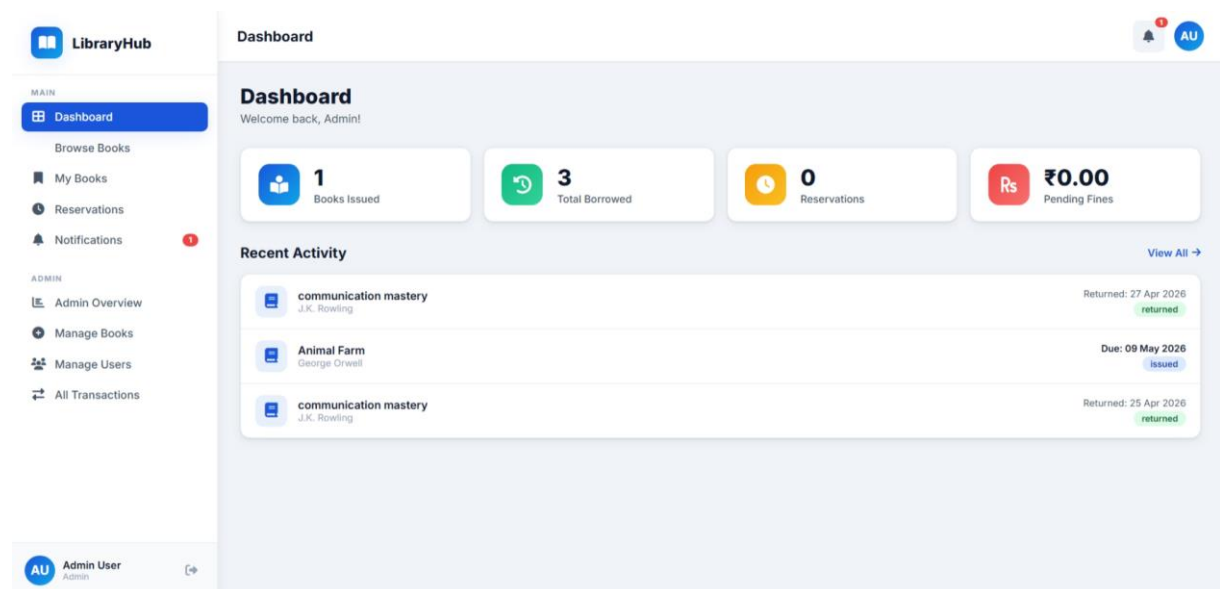
Reserve Book:

- If unavailable → Reserve

Return Book:

- Updates availability

11. PROJECT PROTOTYPE



LibraryHub

MAIN

Dashboard

Browse Books

My Books

Reservations

Notifications

ADMIN

Admin Overview

Manage Books

Manage Users

All Transactions

AUAdmin UserAdmin

Browse Books

Explore our complete collection

Search by title, author, ISBN...

All Genres

All Books

Clear

Available

communication mastery

J.K. Rowling

Fiction

2013 · 113 pages · 1/1 copies

ViewIssue

Available

Harry Potter and the Philosopher Stone

J.K. Rowling

Fantasy

1997 · 223 pages · 3/3 copies

ViewIssue

Unavailable

Animal Farm

George Orwell

Fiction

1945 · 112 pages · 0/2 copies

ViewReserve

Available

1984

George Orwell

Science Fiction

1949 · 328 pages · 2/2 copies

ViewIssue

Available

Murder on the Orient Express

Agatha Christie

Mystery

1934 · 274 pages · 2/2 copies

ViewIssue

Available

Sapiens: A Brief History of Humankind

Yuval Noah Harari

Non-Fiction

2015 · 443 pages · 3/3 copies

ViewIssue

Available

The Alchemist

Paulo Coelho

Fiction

1988 · 197 pages · 3/3 copies

ViewIssue

Available

A Brief History of Time

Stephen Hawking

Science

1988 · 212 pages · 2/2 copies

ViewIssue

Available

Half of a Yellow Sun

Chimamanda Ngozi Adichie

Fiction

2006 · 435 pages · 2/2 copies

ViewIssue

Available

The Da Vinci Code

Dan Brown

Mystery

2003 · 689 pages · 3/3 copies

ViewIssue

Available

Harry Potter and the Chamber of Secrets

J.K. Rowling

Fantasy

1998 · 251 pages · 2/2 copies

ViewIssue

Available

We Should All Be Feminists

Chimamanda Ngozi Adichie

Non-Fiction

2014 · 64 pages · 3/3 copies

ViewIssue

LibraryHub

MAIN

Dashboard

Browse Books

My Books

Reservations

Notifications

ADMIN

Admin Overview

Manage Books

Manage Users

All Transactions

AUAdmin UserAdmin

My Books

Track your issued books and history

Currently Issued

History

Animal Farm

George Orwell

Issued: 25 Apr 2026

Due: 09 May 2026

12 day(s) remaining

IssuedReturn

LibraryHub

MAIN

Dashboard

Browse Books

My Books

Reservations

Notifications1

ADMIN

Admin Overview

Manage Books

Manage Users

All Transactions

AUAdmin UserAdmin

Notifications

Stay updated with your library activity

Mark All Read

Book Issued Successfully

"communication mastery" is due on 2026-05-11. Please return it on time to avoid fines.

12m ago

Book Issued Successfully

"Animal Farm" is due on 2026-05-09. Please return it on time to avoid fines.

1d ago

Book Issued Successfully

"communication mastery" is due on 2026-05-09. Please return it on time to avoid fines.

1d ago

LibraryHub

MAIN

Dashboard

Browse Books

My Books

Reservations

Notifications1

ADMIN

Admin Overview

Manage Books

Manage Users

All Transactions

AUAdmin UserAdmin

Admin Overview

Library statistics and management

12

Total Books

1

Registered Users

2

Active Issues

0

Overdue

0

Reservations

Rs

₹0.00

Unpaid Fines

Send Overdue Notifications

Recent Transactions

BOOK	USER	ISSUE DATE	DUE DATE	STATUS
communication mastery	Admin User	27 Apr 2026	11 May 2026	returned
Animal Farm	Admin User	25 Apr 2026	09 May 2026	issued
communication mastery	Admin User	25 Apr 2026	09 May 2026	returned
Animal Farm	Dolly	20 Apr 2026	04 May 2026	issued

MAIN

- Dashboard
- Browse Books
- My Books
- Reservations
- Notifications 1

ADMIN

- Admin Overview
- Manage Books**
- Manage Users
- All Transactions

AU Admin User Admin

Manage Books

Add, edit, and remove books from the system

+ Add New Book

+ Add Author

+ Add Genre

Search books...

TITLE	AUTHOR	GENRE	ISBN	COPIES	AVAILABLE	ACTIONS
communication mastery	J.K. Rowling	Fiction	456-7-789-33356-7	1	1	
Harry Potter and the Philosopher Stone	J.K. Rowling	Fantasy	978-0-7475-3269-9	3	3	
Animal Farm	George Orwell	Fiction	978-0-452-28424-1	2	0	
1984	George Orwell	Science Fiction	978-0-452-28423-4	2	2	
Murder on the Orient Express	Agatha Christie	Mystery	978-0-00-712036-3	2	2	
Sapiens: A Brief History of Humankind	Yuval Noah Harari	Non-Fiction	978-0-06-231609-7	3	3	
The Alchemist	Paulo Coelho	Fiction	978-0-06-112241-5	3	3	
A Brief History of Time	Stephen Hawking	Science	978-0-553-38016-3	2	2	
Half of a Yellow Sun	Chimamanda Ngozi Adichie	Fiction	978-1-4000-9551-9	2	2	
The Da Vinci Code	Dan Brown	Mystery	978-0-385-50420-5	3	3	
Harry Potter and the Chamber of Secrets	J.K. Rowling	Fantasy	978-0-439-06486-6	2	2	
We Should All Be Feminists	Chimamanda Ngozi Adichie	Non-Fiction	978-1-101-91176-2	3	3	

MAIN

- Dashboard
- Browse Books
- My Books
- Reservations
- Notifications 1

ADMIN

- Admin Overview
- Manage Books
- Manage Users
- All Transactions**

AU Admin User Admin

All Transactions

Complete issue and return history

Currently Issued

Overdue

Returned

All

BOOK	USER	ISSUE DATE	DUE DATE	RETURN DATE	FINE	STATUS	ACTION
Animal Farm George Orwell	Admin User admin@library.com	25 Apr 2026	09 May 2026	—	₹0	Issued	Return
Animal Farm George Orwell	Dolly dolly.sh2006@gmail.com	20 Apr 2026	04 May 2026	—	₹0	Issued	Return

- Browse Books Page
- Reservation Page
- Dashboard
- Login Page

12. CONCLUSION

The Library Management System successfully automates key library operations and improves efficiency. It demonstrates full-stack development using modern technologies and provides a

scalable foundation for future enhancements such as digital libraries and AI-based recommendations.

13. REFERENCES

- [Node.js Documentation](#)
- [Express.js Documentation](#)
- [MySQL Documentation](#)
- [MDN Web Docs](#)